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International Journal of Production Research

ISSN: 0020-7543 (Print) 1366-588X (Online) Journal homepage: www.tandfonline.com/journals/tprs20

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To cite this article: Zhongyang Li, Zhaohui Deng, Zhiguang Ge, Lishu Lv & Jimin Ge (2023) A hybrid approach of case-based reasoning and process reasoning to typical parts grinding process intelligent decision, International Journal of Production Research, 61:2, 503-519, DOI: 10.1080/00207543.2021.2010144

To link to this article: https://doi.org/10.1080/00207543.2021.2010144

Published online: 24 Dec 2021.

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INTERNATIONAL JOURNAL OF PRODUCTION RESEARCH
2023, VOL. 61, NO. 2, 503–519
https://doi.org/10.1080/00207543.2021.2010144

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A hybrid approach of case-based reasoning and process reasoning to typical parts grinding process intelligent decision

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ABSTRACT

After grinding, the machine tool spindle with high surface integrity has a significant impact on its subsequent service life. Therefore, it is necessary to create a reasonable process plan for the grinding process of the machine tool spindle. A hybrid method based on case-based reasoning (CBR) and process reasoning (PR) was proposed in this paper. The subjective and objective weights of feature attributes are determined in the CBR by the AHP method and the CRITIC method, respectively, and the similarities between the latest case and the retrieved case are calculated on the basis of the nearest neighbour algorithm. The most suitable process solution can be selected from the case database by means of CBR. PR is applied to solve the problem that the case cannot be retrieved through CBR or is unsatisfactory. In the case of the grinding machine spindle, the applicability of this technology was demonstrated. Consequently, the results revealed that the surface consistency of the spindle after grinding was substantially increased. A decision-making system based on the proposed approach was developed by using Qt 4.8.7 and SQLite 3. The results demonstrate the viability and efficacy of the hybrid CBR-PR method to rapidly generate process planning for specific parts.

ARTICLE HISTORY
Received 7 May 2021
Accepted 18 November 2021

KEYWORDS
Grinding; process planning; intelligent decision; case-based reasoning; process reasoning

1. Introduction

Typical parts, such as hemispherical parts, automotive camshaft, machine tool spindle, nuclear power valve, moulds, machine tool guide, are key components of manufacturing industries including rail transportation, energy equipment, mould manufacturing, high-end CNC machine equipment and aerospace. And the realisation of high-precision, high-efficiency, compound and intelligent processing of these parts plays an essential role in improving the processing accuracy and processing efficiency of the multi-function CNC grinding machine and reducing the processing cost of the equipment manufacturing industry.

The spindle is one of the core components of the machine tool, which is used in almost all rotary processing machine tools, including grinding machines, lathes, milling machines, boring machines, machining centres, etc. The quality of the spindle directly determines the working accuracy of the machine tool, which requires the spindle to have high precision, good surface quality, and good quality consistency (Hong and Yin 2016; Liu et al. 2020). However, the spindle of the machine tool has the characteristics of high speed, a large force, and easy wear and failure. During its entire processing process, each process will affect the final quality of the spindle (Manikandan and Prabagaran 2021). Grinding is one of its final processing methods. It is an important means for the high precision and high surface quality requirements by reasonably planning the grinding process plan.

Grinding processing is a multi-factor coupled nonlinear machining process (Huang et al. 2021). The machining mechanism is closely related to the characteristics and influence of various factors such as material science, high-speed cutting, friction, lubrication, dynamics, etc., which makes the research on the grinding mechanism becomes more complicated and difficult than the research on cutting mechanism (Singh et al. 2020; Zhu et al. 2020; Zhu and Beaucamp 2020). Due to the complex mechanism of grinding, there are many factors that affect the processing technology, and it is difficult to detect the processing parameters online. Therefore, the optimisation of the process plan and the reduction of processing errors in the grinding process have become problems that trouble the manufacturing enterprises (Zhao et al. 2018; Lv et al. 2020).

In order to optimise computing performance and decision-making process stability for the unique feature attributes of the machine tool spindle grinding process, a method of grinding process planning for machine tool spindle based on CBR and PR is proposed.

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The proposed method calculates the weights of feature attributes by combining subjective and objective weights, and can quickly match suitable process plans. In the proposed system, the cases were stored in the case database based on the case features including process problem descriptions and corresponding processing solutions and application verification. To measure the subjective and objective weights of feature attributes, respectively, AHP and CRITIC were used. And by the linear weighting principle, the comprehensive weights were determined. The resemblance between the aim case and the retrieved cases was evaluated by the 'nearest neighbor algorithm (NNA)' while using CBR for the retrieval of cases. Then the closest matching case was obtained by comprehensive evaluation method. When a suitable case is not retrieved or unsatisfactory, the partially optimised parameters will be obtained through PR. The final processing plan can be obtained by synthesising the results of CBR and PR. Taking the scheduling of the spindle grinding process as an illustration, the efficacy of this approach is tested, which can enhance the surface quality of processed parts and reduce the processing time. Accordingly, a decision-making system based on the proposed approach was also developed.

The remaining of this paper is organised as follows. Relevant literature on CBR and process optimisation method applied in manufacturing was discussed in Section 2. In Section 3, we proposed CBR-PR based intelligent decision system framework for machine tool spindle grinding process and related theoretical realisation. Section 4 used a case study of a grinding wheel spindle to show how the suggested technology would be used in industry. The software framework and interactive application process of the proposed system are demonstrated in Section 5. Section 6 is the summary and conclusion.

2. Literature review

Through searching similar historical cases, case-based reasoning (CBR) uses specific knowledge in existing experience or results to deal with new problems (Aamodt and Plaza 1994). CBR focuses on solving the problems through similar ones (Finnie and Sun 2003). CBR has been broadly applied in many fields such as commerce (Leśniak and Krzysztof 2018), environmental engineering (Zhao et al. 2019), medical diagnosis (Brown et al. 2018), electronic communication (Mohammed et al. 2018), automation (Khosravani, Nasiri, and Weinberg 2019), energy industry (González-Briones et al. 2018), construction industry (Faia et al. 2017), assembly sequence planning (Su 2007), etc. Meantime, CBR also has a large number of application scenarios in the field of processing and manufacturing, including NC code generating system (Ben-Arieh and Chopra 1997), injection mould design (Mok, Kwong, and Lau 2000; Mok, Hua, and Wong 2008). In order to promote information and knowledge exchange between all stakeholders, a new knowledge-based maintenance planning framework was adopted that incorporated case-based logic concepts to increase the effectiveness of maintenance planning (Wan et al. 2019). An ontologically-supported CBR approach was suggested for computer-assisted tolerance requirements (Qin et al. 2018). For remanufacturing process preparation, the combination approach of cased-based reasoning (CBR) and rough set (RS) was proposed. In order to accurately determine the most relevant response from the case database, CBR was used to measure the similarity of deposited cases (Jiang et al. 2019). Furthermore, a new qualitative case-based reasoning and learning algorithm (QCBRL) using CBR approaches to retrieve and replicate cases using qualitatively spatial images to choose the right behaviour in a challenging setting by linking items in an environment in order to maximise the final outcome (Homem et al. 2020). Chougule and Ravi∗ (2005) applied AHP-based NNA for case retrieval in CBR. A comprehensive method of RS and CBR has been proposed to recover the camshaft grinding mechanism, which can be easily and reliably selected from a wide database with various cases, and the computing similitude checking problems have been minimised considerably. Furthermore, a new R^5 model based on (Finnie and Sun 2003) was proposed (Zhang et al. 2013). For further standardise the description, a new R^6 CBR model based on comprehensive subjective and objective weights was proposed in Figure 1, which includes *Representation*, *Reclassification*, *Retrieve*, *Reuse*, *Revise*, *Retain*. From the reviews above, there are few CBR studies on grinding, so how to further achieve effective application in grinding becomes a point of concern.

Although CBR can utilise the previous processing plan to realise processing applications by matching feature attribute similarity, not all the retrieved examples can meet the processing requirements. Taking the machine tool spindle as an example, not only do different types of processing machine tools have different spindles, but the same type of machine tool may also have different spindles, which results in a wide variety of machine tool spindles. Consequently, the processing parameters of these machine tool spindles are different, and it is not always possible to retrieve the most suitable machining process plan only through CBR. Therefore, when CBR is unable to retrieve a suitable case or the retrieved cases are unsatisfactory, the method of process reasoning (PR) can be used to make up for this deficiency.

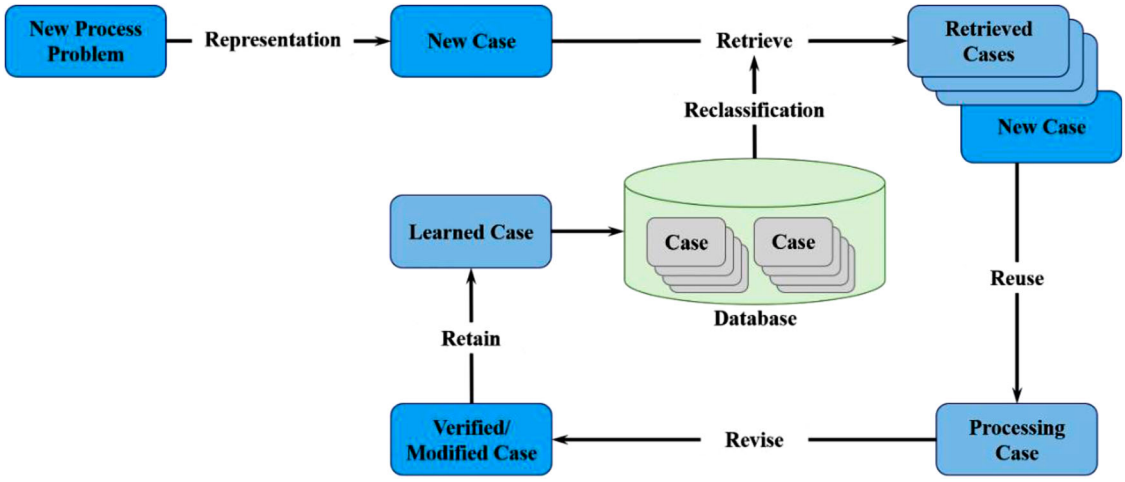


Figure 1. The R^6 model of CBR cycle.

Currently, process parameter optimisation has rich approaches, such as response surface methodology (RSM), genetic algorithm (GA), multi-objective optimisation method (mostly grey relational analysis, GRA) and other creative algorithms. Öztürk and Kahraman (2019) applied RSM and Monte Carlo simulation-based probabilistic uncertainty analysis to model and optimise flat glass grinding processing parameters. RSM has conducted mathematical modelling to establish relationships between process parameters and performance response(s) and has applied multi-objective genetic algorithm (MOGA) to optimise vibration and surface roughness (SR) at the same time (Rudrapati, Pal, and Bandyopadhyay 2016). Sanjeevi, Kumar, and Krishnan (2020) predicted the optimal results of SR using RSM in EN 24 steel surface grinding machining process. With hybrid particle swarm optimisation, a process planning method incorporating the RSM of the evolution of SR for multi-objective optimisation was proposed (Chen et al. 2018). Awale, Vashista, and Khan Yusufzai (2020) optimised the minimum quantity lubrication (MQL) mist parameters to hardened AISI H13 tool steel plunge grinding by using GRA. In the grinding process for reducing total time and expense while optimising output rate and surface quality under both rough and finishing conditions, a lexicographic weighted Tchebycheff method was utilised to get effective PO solutions (Khalilpourazari and Khalilpourazary 2017). On the basis of GA, with developing a model for grinding multi-objective parameter optimisation, the best efficiency and lowest emissions of 45 steel under grinding processing have been obtained (Deng et al. 2016). To optimise a ceramic grinding process for several objectives, a GA code has been created (Agarwal 2016). By using a hybrid technique called GRA-based principal component analysis (PCA), Kumar Yadav and Kumar Singh Yadav (2020) optimised the material removal rate (MRR) and reduce SR during the Ti-6Al-4V electrochemical cut-off grinding. In the internal cylindrical grinding process for 9CrSi instrument steel, Xuan Hung et al. (2019) refined the dressing parameters to minimise SR and increase the MRR using the Taguchi system and GRA. For achieving the optimum scheme of partial grinding parameters, AISI 1045 steel grinding multi-objective optimisation was carried out. (Wang et al. 2020). Li, Wang, and Yang (2002) also used conventional methods to optimise the grinding parameters. Modified ϵ constrained Differential Evolution (Mod-ϵDE) was utilised for the surface grinding process (SGP) issue of restricted optimisation (Rana and Lalwani 2017).

3. CBR-PR based intelligent decision system framework for machine tool spindle grinding process

It is of great significance to formulate suitable processing technology plan for the machining of machine tool spindle. The proposed method makes full use of the advantages of both case reasoning based on subjective and objective weights and process reasoning based on neural network to facilitate the selection of machine tool spindle machining process parameters, including case representation, case retrieval, process reasoning, case evaluation, and case retain. Figure 2 is the flow map of the decision-making system.

In the CBR-PR based system, the process case information in the case base is called to calculate the feature attribute weight. The CRITIC approach is used to produce the objective feature attribute weights while the APH approach the subjective feature attribute weights, and then the comprehensive feature attributes weight is calculated by the linear weighting principle combining both subjective and objective weights. If there are any

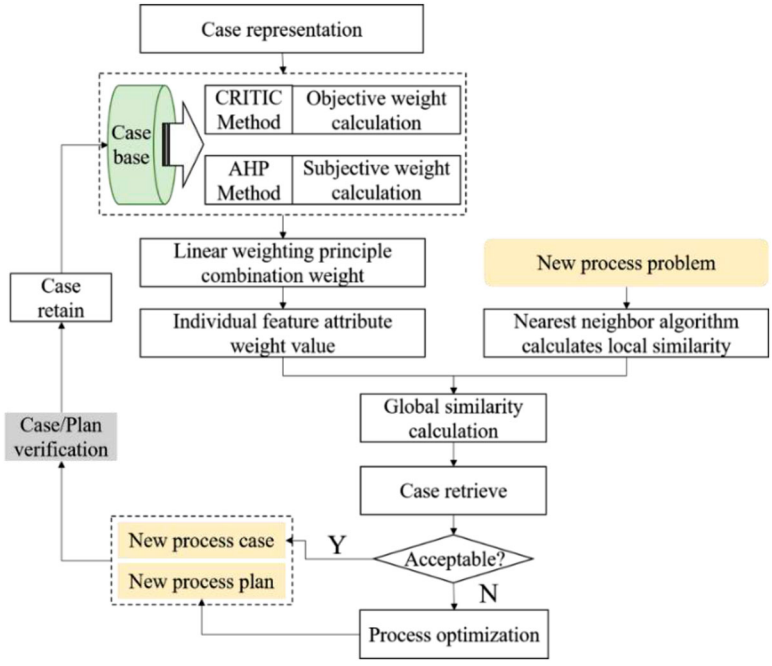


Figure 2. CBR-PR based intelligent decision system framework for machine tool spindle grinding process.

cases in the database that are more closely related to the description of the new process problem by ranking of the global similarity, the NNA can be used in case retrieval to find them. If the selected process case information is not found or unsatisfactory, the process reasoning will be performed on the new process problem, and the final optimal solution will be formed by the results of PR and CBR. Finally, after verifying the retrieved or optimised solution obtained by CBR-PR based system, the revised information and knowledge involved to new case are evaluated and then saved in the case database.

3.1. Case representation

As the most basic part in CBR, case representation enables the computer to complete the identification and transfer of cases with reasonable expression. Case representation has two main characteristics, comprehensively described and concise. Based on the characteristics, this article chooses a framed structured representation method. The complete process example contains the following three parts of information:(1) description of process problems; (2) solutions to process problems; (3) application results and case evaluation. Therefore, the machine tool spindle grinding case can be represented in the form of 'case number + description of the process problems + solutions of the process problems + application results and case evaluation'. The case number is automatically generated by the process software to determine the unique identifier of the case.

3.2. Subjective and objective weight calculation

Machine tool spindle grinding process cases include a variety of feature attributes. Different feature attributes have different effects on the processing results of the process plan. This paper presented a combination weighting approach of subjective and objective weights based on the AHP (Saaty 1977) and CRITIC (Diakoulaki, Mavrotas, and Papayannakis 1995) method for the machine tool spindle grinding process feature attributes. The combination of these two methods not only avoids the inaccuracy of the weight calculation caused by the data deviation in the case data collection process, but also takes the experience value of the domain experts into consideration. This reduces the arbitrariness of the weight assignment, and considers the authenticity of the data, so that the weighting result is true and reliable.

(1) Subjective weights determination

1) Construct the machine tool spindle grinding processing feature attributes comparison judgment matrix $T = [t_{ij}]_{n \times n}$, n is the feature attributes amount. t_{ij} in the matrix is value of the importance of attribute i to attribute j;
2) Consult with experts in the field of CNC spindle grinding of machine tools. This article uses the 1–9 ratio scaling method to define t_{ij}.
3) After normalising each column of the judgment matrix T, the corresponding weight of each feature attribute can be obtained. The calculation formula is

as follows:

$$\omega_{SF_i} = \frac{(\prod_{j=1}^{n} t_{ij})^{\frac{1}{n}}}{\sum_{i=1}^{n} (\prod_{j=1}^{n} t_{ij})^{\frac{1}{n}}} \quad (1)$$

where ω_{SF_i} is the ith feature attribute weight coefficient obtained using the AHP method.

4) The consistency testing. For the purpose of ensuring the validity of a suggested solution, the consistency ratio is used to evaluate the consistency of a judgment matrix built on expert knowledge. The consistency index CI $= \frac{\lambda - n}{n-1}$, λ is the judgment matrix maximum eigenvalue. The random consistency index RI, corresponding to the number of characteristic attributes, is obtained by looking up the table. CR = CI/RI is defined as the consistency ratio, and the consistency testing is passed while CR < 0.1.

(2) Objective weights determination

Build a decision matrix on the collected data A $= [a_{ij}]_{m \times n}$, $i = 1,2,3,\ldots, m$; $j = 1,2,3,\ldots, n$. Among them, m is the process cases amount, and n is the feature attributes amount.

1) Normalise the decision matrix and calculate the contrast intensity s_j:

$$s_j = \sqrt{\frac{\sum_{j=1}^{n} (\overline{y_j} - y_{ij})^2}{m-1}} \quad (2)$$

where $\overline{y_j}$ is the sample mean of the jth feature attribute.

2) Calculate the conflicting c_j:

$$c_j = \sum_{j=1}^{n} (1 - r_{ij}) \quad (3)$$

where r_{ij} is the correlation coefficient between feature attributes, c_j reflects the conflict between the jth and other feature attributes.

3) Calculate objective weight ω_{OF_j}:

$$\omega_{OF_j} = \frac{s_j c_j}{\sum_{j=1}^{n} s_j c_j} \quad (4)$$

(3) Comprehensive weight determination

To make a more appropriate weight value of the machine tool spindle grinding process feature attribute, the weight coefficient calculated by the above two methods is used to calculate the comprehensive weight by linear weighting principle. For a certain feature attribute F_i, its comprehensive weight is

$$\omega_{F_i} = \alpha \omega_{SF_i} + (1 - \alpha)\omega_{OF_i} \quad (5)$$

where ω_{F_i} is the comprehensive weight of feature attribute i; ω_{SF_i} is the weight of feature attribute i obtained using subjective weighting method AHP method; ω_{OF_i} is the weight of feature attribute i obtained using objective weighting method CRITIC method; α is the weighted coefficient, generally 1/2.

3.3. Case retrieval

After the weight value is determined, the global similarity between the target process problem and each case in the database needs to be calculated by the system to select the appropriate one to match. The local similarity of each feature attribute was combined for calculating the global similarity. In this paper, the local similarity of grinding features was determined by using NNA method. The key concept for the NNA algorithm is that if a sample in the functional space is categorised like other k-classified samples, the sample still falls in this category and has sample characteristics in this category. Hence, the NNA approach can be extended to various kinds of values, which satisfies the demand for the diverse characteristic expression of the grinding process. The distance between two values can be measured through the Euclidian and the Manhattan distances. Through Equation (6), the global similarity can be determined.

$$SIM(X, Y) = \sum_{i=1}^{m} (sim(F_i^X, F_i^Y)^l \omega(F_i))^{1/l} \quad (6)$$

where X represent new case, and Y represent deposited case. The subscript i represents the ith feature attribute of the case. $\omega(F_i)$ is the weight of the feature F_i. $SIM(X, Y)$ is the global similarity between case X and case Y. $sim(F_i^X, F_i^Y)$ represents the local similarity between F_i^X and F_i^Y. And l in this article takes the Manhattan distance with the value of 1.

According to the feature attributes described in the machine tool spindle grinding process problem, the case feature attributes are classified according to different types. Different types of feature attributes have different methods for calculating similarity. This article divides the feature attributes into the following three types:

For numeric feature attribute, the local similarity calculation is based on the formula as followed:

$$sim(F_i^X, F_i^Y) = 1 - \frac{|F_i^X - F_i^Y|}{\max(F_i) - \min(F_i)} \quad (7)$$

where $\max(F_i)$ is the maximum value of the feature attribute F_i while $\min(F_i)$ is the minimum value.

Table 1. Allocation of surface burn grade values.

Degree	No	Mild	Serious
F_i	0	0.5	1

Table 2. Local similarity of the material heat treatment method.

Type	Quenching	Normalising	Tempering	Annealing
Quenching	1	0.9	0.8	0.1
Normalising		1	0.9	0.1
Tempering			1	0.1
Annealing				1

In the machine tool spindle grinding, there are many feature attributes are numeric, such as material hardness, total grinding allowance, processing length, processing diameter, dimensional accuracy, roundness, cylindricity and surface roughness.

For enumeration feature attribute, the determination of these feature attribute values strongly relies on the subjective knowledge of human beings, which are generally given based on experience, and among the attribute values have a certain relationship in the attribute domain. Hence, the local similarity calculation is based on the formula as followed:

$$sim(F_i^X, F_i^Y) = 1 - \frac{|F_i^X - F_i^Y|}{M} \tag{8}$$

where M represents the highest value of the feature enumeration F_i. The assignment values of surface burn are shown in Table 1. In addition, the local similarity of some feature attributes needs to be determined based on domain knowledge. For example, the local similarity of the material heat treatment method is shown in Table 2, and its value represents the difference in processing performance between the two material states.

For linguistic feature attribute, there is no connection between the different values of the feature attributes of the attribute domain with this local similarity. Therefore, the local similarity calculation is based on the formula as followed:

$$sim(F_i^X, F_i^Y) = \begin{cases} 1 & F_i^X = F_i^Y \\ 0 & F_i^X \neq F_i^Y \end{cases} \tag{9}$$

Of Equality. (9), if two features are equivalent, the local similarity will be 1; otherwise the local similarity will be 0. Workpiece type, material type, material grade are linguistic feature attributes.

3.4. Process plan reasoning

When suitable cases are not retrieved through case retrieval or the processing accuracy of the retrieved case does not meet the requirements, the process reasoning module will start. In the spindle grinding process, various factors affect the selection of grinding parameters with nonlinear mapping relationships. The neural network model has the ability of nonlinear mapping, which can reasonably select the parameters that affect the grinding process, effectively establish a relationship model between the description of the grinding process problem and the process solution, and finally obtain the optimal grinding parameters of the process through the established model.

After the grinding processing environment is determined, nonlinear modelling is performed only for the processing results produced by the same type of different processing plans in the same processing environment. Just one hidden layer BP neural network is able to achieve random nonlinear mapping without restricting the hidden layer (layer H) nodes amount. After eliminating the process parameters that have little impact on the machining process, five variables of surface burn degree, dimensional accuracy, surface roughness, roundness and cylindricity are used as input, so the number of input layer (layer I) nodes is set to 5. Then, the speed and allowance that have the greatest influence on the machining result are used as the predicted output, so the linear speed and longitudinal feed speed of the grinding wheel for rough, fine and light grinding, the fine grinding allowance and the light grinding allowance are 8 variables as the output. In order to further decrease the mapping output variables amount of the network model and reduce the complexity of the network topology, rough grinding allowance is indirectly calculated by the total grinding allowance, fine grinding allowance and light grinding allowance, so the nodes amount in the output layer (layer O) is set to 8. And these two layers containing the specific parameters are shown in Figure 3.

In order to calculate the output value of each neuron, the transfer function is often used with the input layer. The tangent function is used in this article to convert the neuron values between input and output. The output of hidden layer neurons and output layer neurons are calculated by Equations (10) and (11)

$$f(X_j^H) = \tanh\left(\sum_{i=1}^{r} X_i^I w_{ij}^{IH} - \theta_{ij}^{IH}\right) \tag{10}$$

$$f(X_k^O) = \tanh\left(\sum_{j=1}^{s} X_j^H w_{jk}^{HO} - \theta_{jk}^{HO}\right) \tag{11}$$

where i, j, and k represent the neurons amounts in the layer I, H, O. X_i^I, X_j^H and X_k^O are the input variable value, hidden variable value and output variable value respectively. w_{ij}^{IH} and w_{jk}^{HO} are the binding weights. θ_{ij}^{IH} and θ_{jk}^{HO} are the biases of the jth and kth neurons.

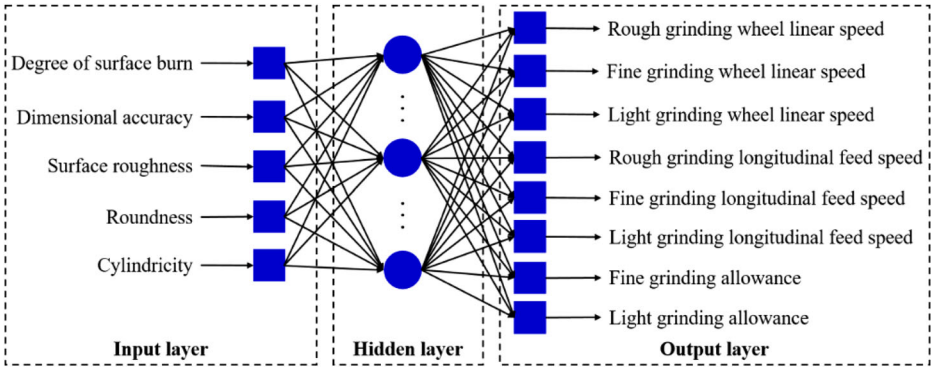


Figure 3. Grinding machine tool spindle neural network model.

The network training times is set to 1000, the learning rate is set to 0.01, the target error is 0.002 and the network training error can be calculated by Equation (12) (Deng et al. 2009)

$$MSE = \frac{1}{m}\sum_{i=1}^{m}(O_i - T_i)^2 \tag{12}$$

where O_i and T_i represent the predicted output value and the network output value for the ith training sample, respectively. And m is the size of the training sample set or the test sample set.

The quantity of node units in the hidden layer determines the neural network recognition accuracy and capability. The amount of hidden layer nodes can be obtained by Equation (13) (Deng et al. 2009)

$$z = \sqrt{n_i + n_0} + a \tag{13}$$

where z, n_i, n_0 are the layer H nodes amount, the layer I nodes amount; and the layer O nodes amount. And a is an integer range 1–10.

According to the analysis above, through programming and comparison optimisation, the hidden layer nodes amount in the network is 10. Consequently, the BP neural network model structure established in this paper is $5 \times 10 \times 8$, as is demonstrated in Figure 3.

If the machine tool spindle grinding neural network model has been developed, it is necessary to use the empirical data to train and test the model in the case database. The neural network model testing samples choose the same kind of process case set in the same processing environment as the parts to be processed and categorise them for training. In order to assess the network's consistency, four early processed data samples are extracted from the case database after the sample training is finished. The test sample values are shown in Table 3, T_1 to T_{13} correspond to the input and output in Figure 3. Data normalisation is used to speed up the convergence process as the value spectrum of certain parameter network input and output vectors is too broad. The neural network training effect is judged by the values of the slope m, intercept b, and agreement R of the regression curve. It is calculated that the slope of the regression curve is $m = 0.9325$, the intercept $b = 0.0269$, and the coincidence degree $R = 0.926633$. After calculation, the MSE of the BP neural network is 5.89%, which is within 10%, so the non-linear mapping of input and output can be well realised.

In addition, the random forest (RF) algorithm and the support vector machine (SVM) algorithm are utilised for comparison. Data from the database, according to the ratio of 8:2, is separated into the training set and test set, and SVM algorithm and RF algorithm are used for training, respectively. The results show that when the SVM algorithm is used, the average MSE of the test set is 13.65% and $R^2 = 0.8215$. The average MSE here is because SVM algorithm only supports a single output, so it is the result of averaging the error of each output variable. When using the RF algorithm, the average MSE of the test set is 16.72%, and $R^2 = 0.7515$. It can be seen that in the multi-input and multi-output nonlinear regression problem like grinding process prediction, the result of using BP neural network is better than SVM and RF. Moreover, better prediction results can be obtained after improving these algorithms, but when the software algorithm programming on the machine tool side is too complicated, it will lead to low running efficiency and even crashes. Therefore, it is feasible to choose BP neural network for process reasoning in this paper.

3.5. Case evaluation

To ensure that the data is suitable for the application, the output of the system needs to be evaluated. The similarity obtained from the data, which represents the matching degree between the target problem and the stored case, has certain objectivity. However, the result is easily affected by the data that may have problems. Confidence has a certain degree of subjectivity because it

Table 3. Test samples of network.

	Input					Output							
No.	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9	T_{10}	T_{11}	T_{12}	T_{13}
1	No burns	0.02	0.5	0.003	0.005	60	65	70	1	0.1	0.05	0.05	0.02
2	No burns	0.02	0.32	0.001	0.002	60	75	80	1.5	0.2	0.1	0.15	0.08
3	No burns	0.05	0.4	0.001	0.004	60	65	70	2	0.3	0.1	0.05	0.08
4	No burns	0.01	0.2	0.002	0.003	70	80	90	1	0.3	0.05	0.05	0.05

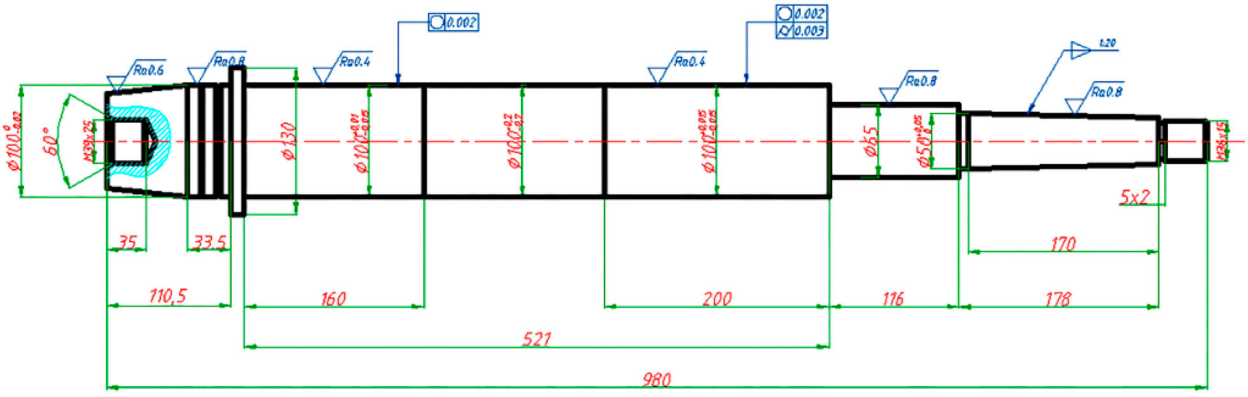


Figure 4. Part drawing of grinder spindle.

is obtained through human judgment. A comprehensive evaluation method that combines similarity and confidence is employed for enhancing the robustness and reliability of case matching. A comprehensive confidence factor E is defined as shown in Equation (14) on the basis of similarity and confidence. (Zhang et al. 2013; Jiang et al. 2019)

$$E(C_i) = \varphi SIM(X, C_i) + (1 - \varphi)\varepsilon(C_i) \qquad (14)$$

where $\varepsilon(C_i)$ represents the confidence of the ith case. φ is the scale factor. The value of scale factor φ (Deng et al. 2010; Li et al. 2020) is generally assigned as 0.7.

After the case is retrieved, the proposed system will automatically calculate the comprehensive confidence E of the selected case. After executing in ascending order, the user will be presented with the best match with the largest E value. After process reasoning, through experimental verification of the inferred parameters, an appropriate confidence will be set for the case, and it will be stored in the database as a resource case for next case retrieving.

4. Application

Taking the grinding wheel spindle of a machine tool factory as an experimental object, the parts drawing is shown in Figure 4. The cylindrical grinding process test of the grinding spindle $\varphi 100 \pm 0.015$ was conducted on the MKG UHS cylindrical grinding machine tool produced from Huda Haijie (Hunan) Engineering Technology Research Co., Ltd. The cylindrical grinding of this grinding machine spindle is determined by 13 condition

Table 4. Feature attribute information table of machine tool spindle.

Case	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}
C_1	Grinding wheel spindle	Alloy steel	65Mn	45	Quenched	0.4	No	105	62	0.020	0.002	0.005	0.4
C_2	Grinding wheel spindle	Cast iron	QT700	52	Quenched	0.4	No	75	60	0.015	0.002	0.002	0.32
C_3	Grinding wheel spindle	Alloy steel	38CrMoAl	65	Quenched	0.4	Mild	100	50	0.015	0.001	0.005	0.32
C_4	Grinding wheel spindle	Alloy steel	38CrMoAl	68	Quenched	0.3	No	100	45	0.025	0.005	0.004	0.4
C_5	Grinding wheel spindle	Alloy steel	65Mn	43	Quenched	0.4	No	168	62	0.015	0.002	0.004	0.32
C_6	Grinding wheel spindle	Alloy steel	38CrMoAl	66	Quenched	0.2	No	200	100	0.018	0.004	0.004	0.8
…	…	…	…	…	…	…	…	…	…	…	…	…	…
C_{71}	Milling machine spindle	Alloy steel	65Mn	50	Quenched	0.5	No	200	120	0.030	0.002	0.005	0.4
C_{72}	Milling machine spindle	Alloy steel	38CrMoAl	69	Quenched	0.5	No	220	120	0.020	0.003	0.003	0.6
C_{73}	Milling machine spindle	Alloy steel	38CrMoAl	68	Quenched	0.3	No	100	40	0.015	0.002	0.003	0.6
…	…	…	…	…	…	…	…	…	…	…	…	…	…
C_{110}	Boring machine spindle	Alloy steel	40Cr	50	Quenched	0.3	Mild	50	36	0.025	0.001	0.003	0.6
C_{111}	Boring machine spindle	Alloy steel	40Cr	52	Quenched	0.3	No	150	60	0.020	0.002	0.003	0.4
…	…	…	…	…	…	…	…	…	…	…	…	…	…
C_{182}	Lathe headstock spindle	Alloy steel	65Mn	45	Quenched	0.4	No	175	80	0.025	0.003	0.002	0.30
C_{183}	Lathe headstock spindle	Alloy steel	65Mn	45	Quenched	0.5	No	175	80	0.030	0.003	0.002	0.32

features, such as workpiece type, material type, material grade, material hardness, heat treatment method, total grinding allowance, the degree of surface burn, the processing length, the processing diameter, the dimensional accuracy, the roundness, the cylindricity, and the surface roughness, which correspond to F_1 to F_{13} in Table 4.

Domain experts will select the reasonable cases for depositing in database by eliminating the unqualified ones after collecting the original information of the feature attributes. The machine tool spindle grinding process plans in Huda Haijie (Hunan) Engineering Technology Research Co., Ltd was summarised in 183 cases for four types of machine tool spindle. Table 5 is a minor part of the cases for the machine tool spindle. The case number is shown in the row, and the feature attributes of cases are shown in the columns.

By consulting the spindle grinding experts and describing the characteristic attributes of the machine tool spindle grinding process problems, a judgment matrix for comparing the feature attributes of the machine tool spindle grinding is constructed as followed.

$$T = \begin{bmatrix} 1 & 1 & 1 & 1/5 & 1/2 & 1/5 & 1/3 & 1/8 & 1/8 & 1/6 & 1/9 & 1/8 & 1/9 \\ 1 & 1 & 1 & 1/5 & 1/2 & 1/5 & 1/3 & 1/8 & 1/8 & 1/6 & 1/9 & 1/8 & 1/9 \\ 1 & 1 & 1 & 1/5 & 1/2 & 1/5 & 1/3 & 1/8 & 1/8 & 1/6 & 1/9 & 1/8 & 1/9 \\ 5 & 5 & 5 & 1 & 4 & 1 & 3 & 1/4 & 1/4 & 1/2 & 1/5 & 1/4 & 1/5 \\ 2 & 2 & 2 & 1/4 & 1 & 1/4 & 1/2 & 1/7 & 1/7 & 1/5 & 1/8 & 1/7 & 1/8 \\ 5 & 5 & 5 & 1 & 4 & 1 & 3 & 1/4 & 1/4 & 1/2 & 1/5 & 1/4 & 1/5 \\ 3 & 3 & 3 & 1/3 & 2 & 1/3 & 1 & 1/6 & 1/6 & 1/4 & 1/7 & 1/6 & 1/7 \\ 8 & 8 & 8 & 4 & 7 & 4 & 6 & 1 & 1 & 3 & 1/2 & 1 & 1/2 \\ 8 & 8 & 8 & 4 & 7 & 4 & 6 & 1 & 1 & 3 & 1/2 & 1 & 1/2 \\ 6 & 6 & 6 & 2 & 5 & 2 & 4 & 1/3 & 1/3 & 1 & 1/4 & 1/3 & 1/4 \\ 9 & 9 & 9 & 5 & 8 & 5 & 7 & 2 & 2 & 4 & 1 & 2 & 1 \\ 8 & 8 & 8 & 4 & 7 & 4 & 6 & 1 & 1 & 3 & 1/2 & 1 & 1/2 \\ 9 & 9 & 9 & 5 & 8 & 5 & 7 & 2 & 2 & 4 & 1 & 2 & 1 \end{bmatrix} \tag{15}$$

Through calculation, the maximum eigenvalue of Equation (15) is 13.5801. Therefore, the consistency index CI is 0.048. By looking up the table, when the characteristic attribute value is 13, the random consistency index RI is 1.56. Finally, the value of the consistency ratio CI/RI is 0.0308., The judgment matrix passed the consistency test owing to the consistency ratio is less than 0.1.

The data in the matrix above is normalised by columns and substituted into Equation (1), and the subjective weight values of each characteristic attribute are calculated by the AHP method. Taking the 13th feature attribute surface roughness as an example, the subjective weight is calculated as follows

$$\omega_{SF_{13}} = \frac{(\prod_{j=1}^{13} t_{13,j})^{\frac{1}{13}}}{\sum_{i=1}^{13} (\prod_{j=1}^{13} t_{13,j})^{\frac{1}{13}}} = 0.187$$

The subjective weights of remaining feature attributes are calculated in the same way: $\omega_{SF_1} = 0.013$, $\omega_{SF_2} = 0.013$, $\omega_{SF_3} = 0.013$, $\omega_{SF_4} = 0.046$, $\omega_{SF_5} = 0.018$, $\omega_{SF_6} = 0.046$, $\omega_{SF_7} = 0.025$, $\omega_{SF_8} = 0.129$, $\omega_{SF_9} = 0.129$, $\omega_{SF_{10}} = 0.065$, $\omega_{SF_{11}} = 0.187$, $\omega_{SF_{12}} = 0.129$.

The data in the machine tool spindle grinding process case database is constructed into a decision matrix. After the decision matrix is standardised, the corresponding objective weights can be calculated by CRITIC method according to Equations (2)–(4). Taking the 13th feature attribute surface roughness as an example, the objective weight is calculated as follows

$$\omega_{OF_{13}} = \frac{s_{13} c_{13}}{\sum_{j=1}^{13} s_{13} c_{13}} = 0.190$$

The objective weights of remaining feature attributes are calculated in the same way: $\omega_{OF_1} = 0$, $\omega_{OF_2} = 0$, $\omega_{OF_3} = 0$, $\omega_{OF_4} = 0.053$, $\omega_{OF_5} = 0$, $\omega_{OF_6} = 0.053$, $\omega_{OF_7} = 0.028$, $\omega_{OF_8} = 0.138$, $\omega_{OF_9} = 0.138$, $\omega_{OF_{10}} = 0.072$, $\omega_{OF_{11}} = 0.190$, $\omega_{OF_{12}} = 0.138$.

Synthesise the subjective and objective weight values, and calculate the comprehensive weight value of each feature attribute according to Equation (5). Taking the 13th feature attribute surface roughness as an example, the comprehensive weight is calculated as follows

$$\begin{aligned} \omega_{F_{13}} &= 1/2\omega_{SF_{13}} + (1 - 1/2)\omega_{OF_{13}} \\ &= 0.5 \times 0.187 + 0.5 \times 0.190 = 0.188 \end{aligned}$$

The comprehensive weights of remaining feature attributes are calculated in the same way: $\omega_{F_1} = 0.007$, $\omega_{F_2} = 0.007$, $\omega_{F_3} = 0.007$, $\omega_{F_4} = 0.050$, $\omega_{F_5} = 0.009$, $\omega_{F_6} = 0.050$, $\omega_{F_7} = 0.026$, $\omega_{F_8} = 0.133$, $\omega_{F_9} = 0.133$, $\omega_{F_{10}} = 0.069$, $\omega_{F_{11}} = 0.188$, $\omega_{F_{12}} = 0.133$.

According to the part drawing and the known processing requirements, the description of the new process problem of the spindle combined with the standardised representation method of the process information of the part to be processed is shown in Table 5.

After the new process problem is identified, the local similarity is calculated first for each feature, and then Equation (6) will be utilised to estimate the global similarity.

In the grinding process of the grinder spindle, material hardness, total grinding allowance, processing length,

Table 5. The new problem X information description.

	Workpiece type	Material category	Material grade	Material hardness (HRC)	Heat treatment method	Total grinding allowance	Degree of surface burn
X	Grinding wheel spindle	Alloy steel	38CrMoAl	65	Quenched	0.3	No burns
	Processing length	Processing diameter	Dimension accuracy	Roundness	Cylindricity	Surface roughness	
	200	100	0.030	0.002	0.003	0.4	

Table 6. The result of local similarity calculation.

	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}
C_{16}	Grinding wheel spindle	Alloy steel	38CrMoAl	65	Quenched	0.45	No burns	175	84	0.030	0.003	0.003	0.48
X	Grinding wheel spindle	Alloy steel	38CrMoAl	65	Quenched	0.3	No burns	200	100	0.030	0.002	0.003	0.4
Local similarity	1	1	1	1	1	0.936	1	0.919	0.905	1	0.875	1	0.857

processing diameter, dimensional accuracy, roundness, cylindricity and SR are numerical condition features obtained through Equation (7). Degree of surface burn is calculated by Equation (8) while the local similarity of heat treatment method obtained from Table 2. Workpiece type, material type, and material grade are linguistically obtained by using Equation (9). As example, some feature attributes such as workpiece type, total grinding allowance, degree of surface burn and heat treatment method, whose local similarity can be calculated or obtained as below:

$$F_1^X = F_1^{C_{16}} =' \text{ Grinding wheel spindle}',$$
$$sim(F_1^X, F_1^{C_{16}}) = 1$$
$$sim(F_6^X, F_6^{C_{16}}) = 1 - \frac{|0.3 - 0.45|}{2.5 - 0.15} = 0.936$$
$$sim(F_7^X, F_7^{C_{16}}) = 1 - \frac{|0 - 0|}{1} = 1$$
$$sim(F_5^X, F_5^{C_{16}}) = 1$$

Using the same method, the local similarity of other feature attributes can be calculated. The local similarity between the new process problem and the feature attributes of source case C_{16} is presented in Table 6. The detailed information of C_{16} is shown in Table 7. Subsequently, Equation (6) is used for determining the global similarity between the two cases.

Furthermore, the global similarity between deposited case C_{16} and the new process problem can use Equation (6) for calculating as followed

$$\begin{aligned} SIM(X, C_{16}) &= \sum_{i=1}^{13} (sim(F_i^X, F_i^{C_{16}}) \cdot \omega(F_i)) \\ &= sim(F_1^X, F_1^{C_{16}}) \cdot \omega(F_1) + sim(F_2^X, F_2^{C_{16}}) \cdot \omega(F_2) + \cdots + sim(F_{13}^X, F_{13}^{C_{16}}) \cdot \omega(F_{13}) \\ &= 1 \times 0.007 + 1 \times 0.007 + \cdots \\ &\quad + 0.857 \times 0.188 \\ &= 0.923. \end{aligned}$$

The confidence of the case C_{16} is 1.0. The scale factor φ is 0.7. Through Equation (11), the comprehensive confidence factor $E(C_{16})$ is calculated. Eventually, C_{16} is the target case by retrieving in the database.

$$\begin{aligned} E(C_{16}) &= \varphi SIM(X, C_{16}) + (1 - \varphi)\varepsilon(C_{16}) \\ &= 0.7 \times 0.923 + (1 - 0.7) \times 1 = 0.9461. \end{aligned}$$

According to the above steps, the threshold for case retrieval is assigned as 0.90, and then other satisfactory cases can also be retrieved. Eventually, the grinding process planning of the grinder spindle, which is the best suitable one to the target case, would be given based on C_{16}.

However, through experience and analysis, it is found that in this case, the roundness and surface roughness of the grinding machine spindle have not yet met the requirements of the new case, and further process reasoning is required. Through the model established in Figure 3, the corresponding data in Table 4 is used as input, and after network learning, the corresponding output is obtained. Combining the results of case retrieval and process reasoning, the final processing plan is shown in Table 8. Using the processing plan obtained by CBR-PR system decision, the grinding experiment of 10 grinder spindle blanks was carried out. The actual accuracy measurement results and efficiency of the batch of parts after processing are shown in Table 9. After testing, the grinding samples not only showed no grinding burn on the surface, but also obtained better processing results. Therefore, without further manual adjustment, this case can be explicitly used as a processing plan for the new issue. The enterprise application report demonstrated

Table 7. Detailed information of solution C_{16}.

Case Number: C_{16}					
Description of process problems		Solutions to process problems		Efficiency and case evaluation	
workpiece type	Grinding wheel spindle	machine model	MKG1320	Efficiency (min)	60
material category	alloy steel	grinding wheel type	CBN	Confidence	1.0
material grade	38CrMoAl	grinding fluid grade	SCC730A		
material hardness (HRC)	65	grinding fluid type	Water-based		
heat treatment method	Quenched	workpiece speed (r/min)	320		
total grinding allowance (mm)	0.45	feed speed of table (m/min)	0.6		
degree of surface burn	No burns	rough grinding wheel linear speed (m/s)	55		
processing length (mm)	175	fine grinding wheel linear speed (m/s)	65	Additional information	Huda Science and Technology Park, No.186 Guyuan Road, Yuelu District, Changsha, Hunan Province, Huda Haijie (Hunan) Engineering Technology Research Co., Ltd
processing diameter (mm)	84	light grinding wheel linear speed (m/s)	75		
roundness (mm)	0.003	rough grinding longitudinal feed speed (mm/min)	1		
dimensional accuracy (mm)	0.030	fine grinding longitudinal feed speed (mm/min)	0.12		
cylindricity (mm)	0.003	light grinding longitudinal feed speed (mm/min)	0.08		
surface roughness (Ra) (μm)	0.480	rough grinding allowance (mm)	0.33		
		fine grinding allowance (mm)	0.1		
		light grinding allowance (mm)	0.02		
		supply liquid pressure (MPa)	3		
		supply liquid flow rate (L/min)	50		

Table 8. Final process plan for grinding process of the grinder spindle.

Attribute	Value
Machine model	MKG1320
Grinding wheel type	CBN
Grinding fluid grade	SCC730A
Workpiece speed (r / min)	320
Feed speed of table (m / min)	0.6
Rough grinding wheel linear speed (m/s)	60
Fine grinding wheel linear speed (m/s)	70
Light grinding wheel linear speed (m/s)	80
Rough grinding longitudinal feed speed (mm/min)	1
Fine grinding longitudinal feed speed (mm/min)	0.1
Light grinding longitudinal feed speed (mm/min)	0.05
Rough grinding allowance (mm)	0.2
Fine grinding allowance (mm)	0.08
Light grinding allowance (mm)	0.02
Grinding fluid type	Water-based
Supply liquid pressure (MPa)	3
Supply liquid flow rate (L/min)	50

that after the grinding machine spindle is processed by the method proposed in this article, the surface roughness is reduced from the Ra0.8 μm (traditional method) to Ra0.306 μm (hybrid CBR-PR method), which means proposed approach can improve the surface quality of the spindle by up to 61.75%.

We contrasted the proposed method with a strategy based on a rough set and case-based reasoning (RS-CBR) to ensure it was effective. Through the RS-CBR method, in the attribute reduction stage, the total grinding allowance, machining length and machining radius fall into the negative domain, which means that these three characteristic attributes are unnecessary in decision-making. Therefore, the weight of these three feature attributes is directly 0. However, after the comprehensive calculation of the subjective and objective weights in this article, the weights of these three feature attributes are not 0. This is because considering the deviation of the sample data, the expert knowledge in the grinding field is taken into account, resulting in the calculated weight value is closer to the actual decision weight. In the case retrieval stage, due to the different weight calculation results, the case retrieved by the RS-CBR method is C_{48}. After processing through this case, the roundness is 0.004 and the surface roughness is 0.52 μm, which also fails to meet the processing requirements. Because there is no process optimisation module, the RS-CBR method cannot automatically optimise the process plan. At this time, workers are required to participate in the optimisation plan. In summary, no matter from the calculation of feature attribute weights and the feedback to the process results, the CBR-PR system proposed in this paper can more effectively realise the intelligent decision-making of the process plan, and greatly reduce human intervention and improve the processing effectiveness.

Additionally, the enterprise application report also indicated that the proposed system can reduce the

Table 9. Test results of spindle grinding process.

	Surface roughness (Ra) (μm)	Dimensional accuracy(mm)	Roundness (mm)	Cylindricity (mm)	Efficiency (min)	Degree of surface burn	Feasible?
X	0.400	0.015	0.002	0.003	60	No burns	
1	0.306	0.009	0.002	0.003	50	No burns	YES
2	0.316	0.003	0.001	0.002	80	No burns	YES
3	0.352	0.014	0.002	0.003	78	No burns	YES
4	0.345	0.012	0.001	0.003	53	No burns	YES
5	0.368	0.006	0.001	0.002	65	No burns	YES
6	0.321	0.004	0.001	0.002	33	No burns	YES
7	0.372	0.004	0.001	0.002	55	No burns	YES
8	0.350	0.009	0.001	0.002	34	No burns	YES
9	0.371	0.003	0.002	0.003	47	No burns	YES
10	0.330	0.011	0.001	0.002	45	No burns	YES

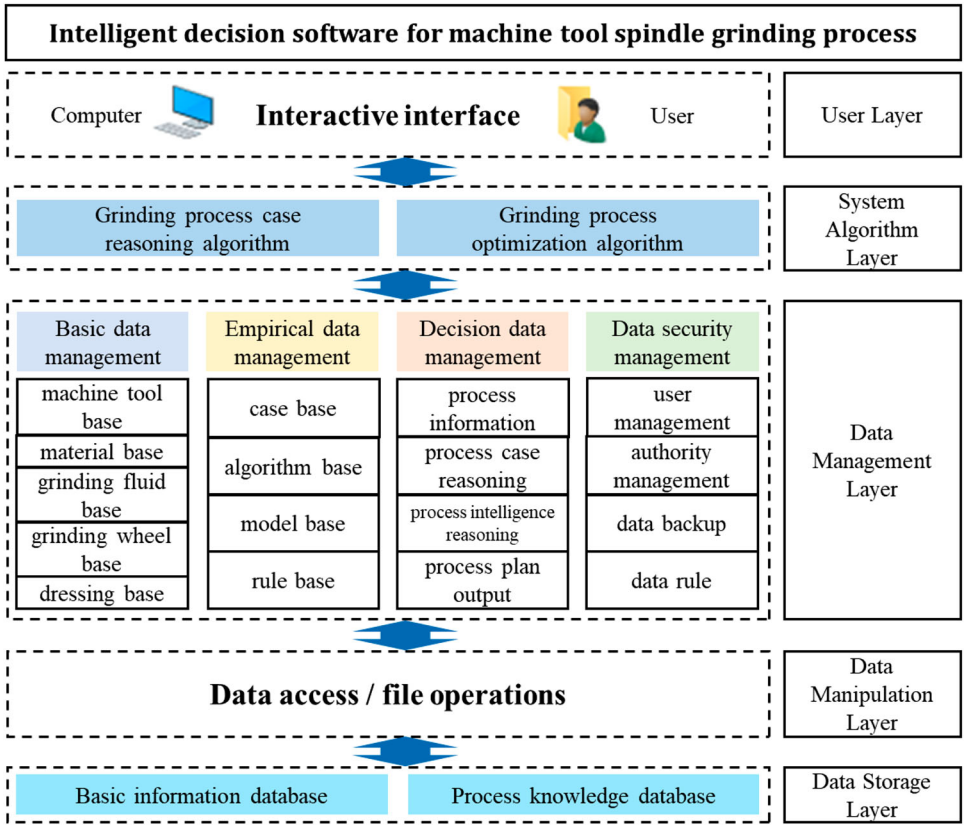


Figure 5. The framework of intelligent decision software for grinding process of machine tool spindle.

decision-making time from nearly 20 h to 2–4 h, and the processing time from 10 min to 5 min. The processing efficiency of a single product is increased by more than 20%, up to 35%. And the decision-making accuracy rate reached 96.8%, giving full advantage to the equipment's process manufacturing capabilities and achieving energy-saving effect. Hence, the results from Tables 8 and 9 show that the proposed CBR-PR system can be used to obtain a grinding process of the grinder spindle through the existing cases. Furthermore, the proposed method can improve the accuracy of case retrieval and generate a better high-precision and efficient grinding process plan for the grinder spindle.

5. Software platform and interaction page

Based on Qt 4.8.7 and SQLite 3, the interactive interface of the grinder spindle technology programme is developed, including the user layer, the system algorithm layer, the data management layer, the data manipulation layer and the data storage layer. Figure 5 is the frame of intelligent decision software for grinding process of machine tool spindle. The specific functions of these layers are as follows.

- *Data storage layer.* This layer stores the basic data and process experience knowledge required by the

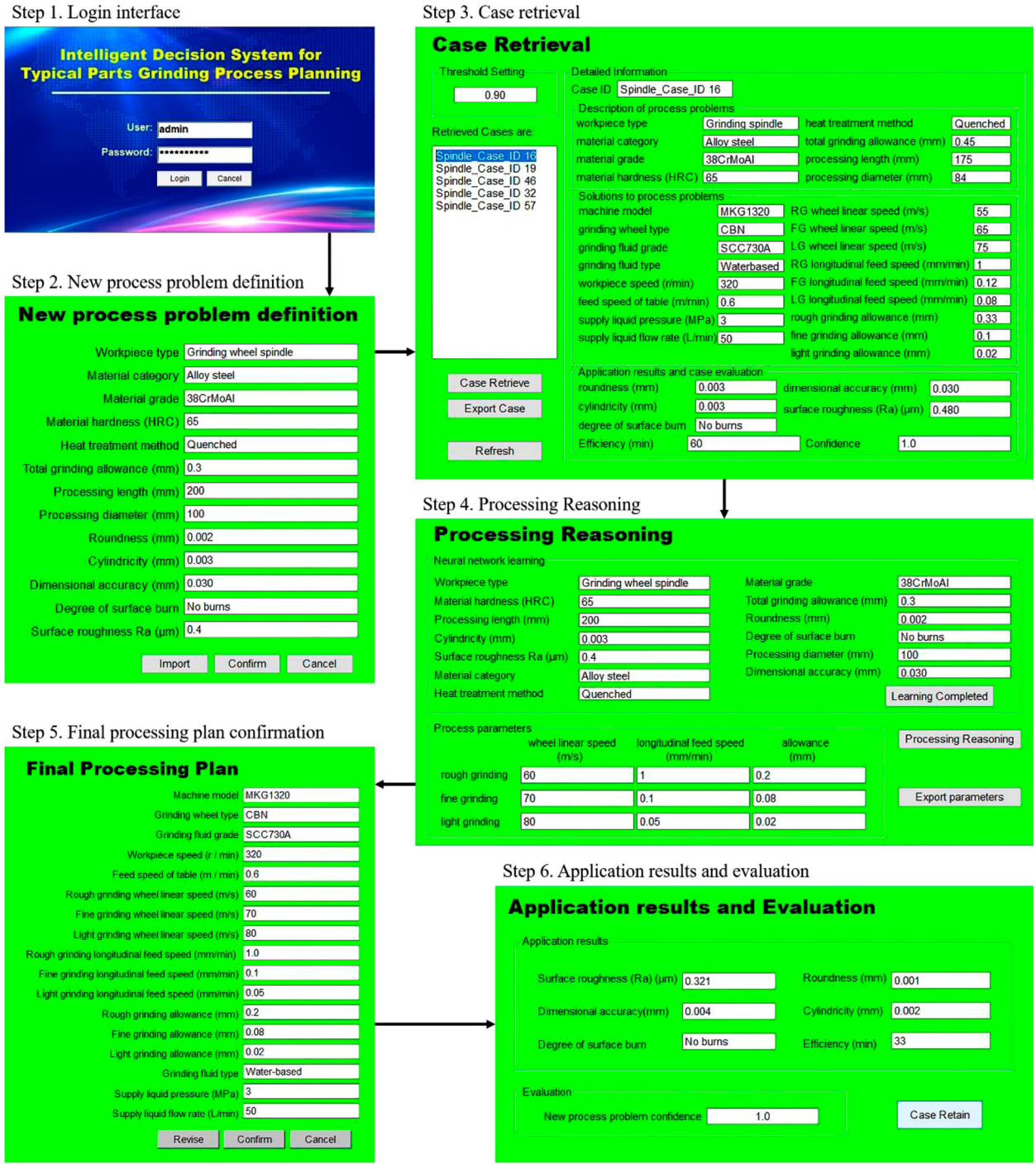


Figure 6. The flow chart of the CBR-PR intelligent decision system.

machine tool spindle grinding process intelligent decision-making software, which is mainly collected through the following three ways: production workshop data, literature data and processing experiment data. The collected data must be audited before it can be stored in the database to ensure the accuracy of the data.

- *Data manipulation layer.* This layer uses data access technology to call the data stored in the data and serves the data management layer.
- *Data management layer.* When used in conjunction with intelligent decision-making software for machine tool spindle grinding, this layer demonstrates most of the software's capabilities and has thus become

essential to the process of software development. And it includes basic data management, experience data management, decision data management and data security management.

- *System algorithm layer.* This layer is an essential component of the intelligent decision-making software for the spindle grinding process of the machine tool. It uses case reasoning algorithms and process reasoning algorithms to provide algorithm support for the output of results in the data management layer.
- *User layer.* This layer is the interface used by database managers or enterprise technical personnel for the process software. The design of this software allows users to operate the process software in the form of a human–computer interaction interface.

The input of process problem information concerning workpiece type, material category, material grade, material hardness (HRC), heat treatment method, etc. or directly import new process problem in text can be realised through this interface. Subsequently, the proposed system will carry out the global similarity calculations, and users can obtain the reference cases with high to low threshold from the existing case database.

Figure 6 is the flow chart of the CBR-PR decision system, which takes the grinding process of the machine tool spindle as an example. After logging in, new process problems can be defined. In the case retrieval module, the staff can retrieve the old cases from the database with the greatest global similarity to the new process problem by setting an appropriate threshold, and view all the process information of the case through the interactive interface. If a suitable case is not retrieved or the retrieved cases are unsatisfactory, a new process parameter can be obtained through the processing reasoning module. Eventually, the final processing plan is given by combining the results of case retrieval and processing reasoning. By processing applications and testing the surface quality and accuracy of the workpiece after processing, new cases that meet the requirements will be evaluated and stored in the database for the coming use.

6. Summary and conclusion

Grinding processing is a multi-factor coupled nonlinear machining process. Determining parts grinding processing plans has traditionally always relied on the experience and knowledge of the workers, which is labour-intensive and error-prone. In this paper, CBR and PR combined method are used to optimise the grinding process planning of typical parts. Through the application results, this method gives a reasonable grinding process planning, and meanwhile improves the efficiency, accuracy and reliability of grinding. In proposed method, the subjective and objective weights of feature attributes are calculated by the AHP method and the CRITIC method, respectively, and the NNA is used to compare the latest case to the one that was retrieved. the similarity between the latest case and the retrieved case is calculated based on the NNA. Through CBR, the most suitable process solution can be selected from the case database. PR is applied to solve the problem that the case cannot be retrieved through CBR or is unsatisfactory. Through retrieving the deposited cases from the case database containing the detailed process planning information of typical parts, the suitable grinding process plan can be determined quickly and effectively. The framework of typical parts grinding process intelligent decision system based on the combination of CBR and PR was built, and the application interactive interface of the system is developed based on Qt 4.8.7 and SQLite 3 programming.

The usefulness and practicability of the approach are demonstrated by the grinder spindle as an illustration. The grinding results obtained from the process plan generated by the new method shows that the parts after processing based on the system decision have the equivalent or even decent quality than the new process requirements which proved that CBR-PR can effective generate reasonable and reliable process plan. There have been successful applications of the process plan suggested in this work to parts such as camshafts and blade tenons. Due to the limitations of the proposed method, the accuracy of the solution is easily affected by the size of the database, parts that are just similar to the processed parts stored in the database can be retrieved for suitable cases, the expert's information is uncertain and inaccurate, and only offline decisions can be made. Future research is to establish a reasonable database size, apply a variety of optimal intelligent algorithms, and develop an intelligent decision cloud platform to enhance the efficiency of case retrieval and achieve intelligent online decision-making for grinding. Moreover, the application scenarios of this method can also be expanded through Monte Carlo simulation.

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Special Fund for the Construction of Hunan Innovative Province [grant number 2020GK2003], National Natural Science Foundation of China [grant number. U1809221] and CZT National Independent Innovation Demonstration Zone Special Project [grant number 2017XK2302].

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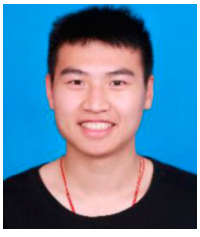
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